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LINGOLY-TOO: DISENTANGLING REASONING FROM KNOWLEDGE WITH TEMPLATISED ORTHOGRAPHIC OBFUSCATION

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ABSTRACT

Frontier language models demonstrate increasing ability at solving reasoning problems, but their performance is often inflated by circumventing reasoning and instead relying on their expanding knowledge and memorisation capacity. We introduce LINGOLY-TOO, a challenging reasoning benchmark of 1,203 questions and a total of 6,995 sub-questions that counters these shortcuts by applying expert-designed obfuscations to Linguistics Olympiad problems. These obfuscations preserve the underlying solution logic while reducing the likelihood problems are solvable with via knowledge or memorisation. Our experiments show that models exploit shortcuts on the original question as performance markedly drop upon obfuscation. Even the best reasoning models remain highly sensitive, with scores dropping from around 0.59 on original problems to 0.48 after obfuscation. LINGOLY-TOO disentangles reasoning from knowledge, offering a clearer measure of true reasoning capabilities.

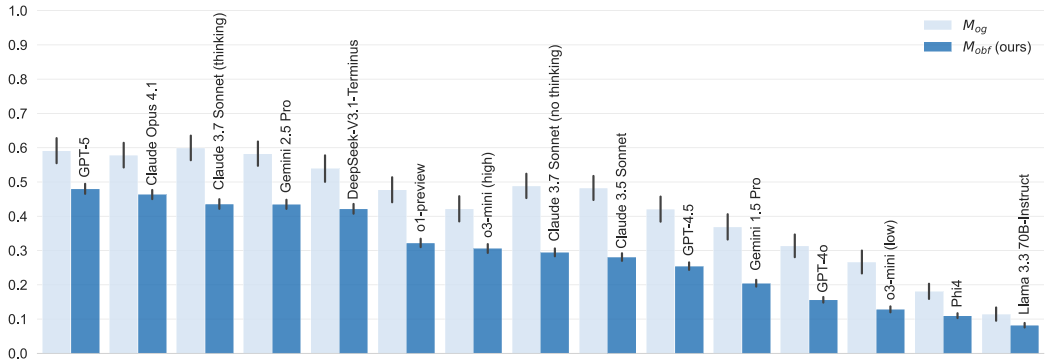

Figure 1: **Reasoning performance on LINGOLY-TOO.** Results without controlling for knowledge and memorisation abilities overestimate reasoning abilities (light blue). Obfuscation mitigates this effect and offers improved estimates (dark blue). There are two scores per model: M_{og} is based on the original problems and M_{obf} is based on obfuscated problems.

1 INTRODUCTION

Experts often define reasoning as the ability to apply abstract rules to derive novel judgments (Koralus, 2022; Huang & Chang, 2023; Lampinen et al., 2024). This ability includes inductive, spatial, causal,

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and commonsense reasoning skills (Kazemi et al., 2025). Benchmarks aim to measure reasoning by testing models on unseen tasks. For such measurements to be valid, reasoning must be both a necessary and sufficient condition for success. However, as training sets grow in size and models' memorisation capacity increases, distinctions between train and test sets, and in- and out-of-domain tasks are blurred (Raji et al., 2021), leading to bias in benchmark estimates.

We distinguish two capabilities resulting from training LLMs that help models solve linguistic tasks. We refer to *knowledge* as information stored in model parameters which captures linguistic, factual, and commonsense patterns useful for downstream tasks. Stored knowledge, while generally useful, can confound the measurement of symbolic reasoning capabilities. For example, GPT-5 can already translate from Welsh to English, hence a linguistic reasoning task based on deriving rules from paired Welsh–English translations would be trivial. We refer to *memorisation* as when models exploit prior exposure to evaluation datasets and report answers based on previously seen problems in training (Li & Flanigan, 2024; Zhou et al., 2023).

Consequently, and amidst the rapid saturation in reasoning benchmarks (Kazemi et al., 2025; Phan et al., 2025), novel methods for generating challenging, unbiased benchmarks are paramount for measuring true reasoning abilities. To prevent models from bypassing reasoning, test cases should minimise the chance that problems are answerable using knowledge or memorisation. Consider the following scenario: a user asks a large language model (LLM) to solve a language puzzle, but it is written in an unfamiliar script or orthography.[1] In this situation, the model's knowledge of the language is rendered useless. The model must deduce grammatical rules and patterns from context, then apply them to solve the given puzzle. Prior work has explored using synthetic test cases (Saparov & He, 2023) to evaluate reasoning. Similar interventions have also been applied to logic puzzles and mathematics, e.g. using symbolic templates to permute variable values (Mirzadeh et al., 2024). However, these small-scale perturbations are less effective because questions remain similar to examples the model may have previously encountered. By contrast, permuting the orthography of a language results in test cases with minimal chance of appearing in training corpus, while the underlying solution logic remains intact.

Our benchmark LINGOLY-TOO operationalises this idea. Original problems are taken from the UK Linguistics Olympiad (UKLO (United Kingdom Linguistics Olympiad, 2023)), which can be solved by high-school students without specific linguistic or domain knowledge. The problems undergo obfuscation: linguistically-informed grapheme-level permutations that preserve the underlying solution logic, remove clues a model could match to prior knowledge, and provide thousands of distinct, solvable variants. Our benchmark aims to test whether a model can use symbolic inductive reasoning to infer the abstract rules and patterns needed to answer the questions from context.

Our evaluation results show that models often rely on language knowledge rather than reasoning, inflating apparent ability, especially in high-resource languages. Although reasoning models, trained to leverage inference-time compute, outperform general-purpose LLMs, they remain sensitive to permutations and exhibit failures such as inconsistent reasoning loops. LINGOLY-TOO addresses key issues in benchmarking LLMs: saturation, contamination, and weak construct validity (Reuel-Lamparth et al., 2024). Our primary contributions are as follows:

- **An unsaturated benchmark for frontier reasoning models.** The top language model on our benchmark, GPT-5, scores 48% overall and only 31% on the highest difficulty problems (Section 4.1).
- **A method to quantify knowledge effects.** The difference between performance on the unperturbed and perturbed problems highlights reasoning shortcuts. We show score inflation from knowledge is correlated with language resourcedness (Section 4.2), and that providing the correct reasoning logic mitigates it (Section 4.3).
- **A method for generating new uncontaminated reasoning problems.** Experiments with then-unpublished UKLO 2025 problems show that perturbation-related performance drops persist, indicating that the effects are not due solely to training-set overlap (Section 4.4).

[1] System for writing a language. A glossary of linguistic terms used in this article is included in Appendix A.

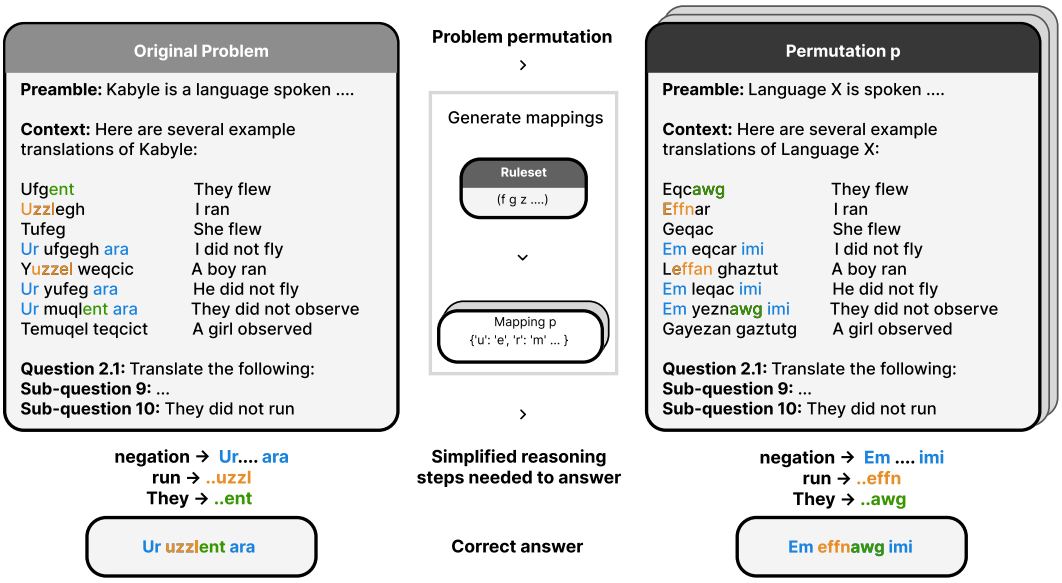

Figure 2: **Obfuscation example.** Problem 160 (Kabyle) before (left) and after (right) obfuscation with the simplified inductive reasoning steps needed for answering sub-question 10. For each obfuscation, we sample a character mapping (permutation) based on the ruleset (see Appendix B.3 to learn more about rulesets), then apply it to obfuscate the problem and the answer. Only the original problem can be solved with the aid of model's internalised knowledge.

2 Related Work

2.1 Reasoning in LLMs

LLMs show rapid progress on task-reasoning across domains, such as mathematical word problems (Cobbe et al., 2021; Hendrycks et al., 2021; Chen et al., 2023; Shao et al., 2024), visual pattern-matching (Chollet, 2019), multi-step planning (Valmeekam et al., 2022; Kambhampati, 2024), and commonsense question-answering (Zelikman et al., 2022; Lin et al., 2020; Rajani et al., 2019; Talmor et al., 2019). It remains unclear to what degree these abilities generalise to novel domains and tasks. Models fail in systematic ways, showing distractability (Shi et al., 2023), content effects (Lampinen et al., 2024), and sharp drops under minor perturbations (Mirzadeh et al., 2024). They also rely on superficial patterns (token bias) (Jiang et al., 2024), and this reliance persists despite inference-time scaling (McCoy et al., 2024). This is further complicated by dataset contamination, where benchmark questions appear in training data (Zhou et al., 2023; Jacovi et al., 2023; Yang et al., 2023). The evidence suggests that high LLM benchmark scores are inflated through shortcuts (Bean et al., 2024).

2.2 Linguistic Reasoning with LLMs

Linguistic reasoning problems test the in-context learning and reasoning ability of LLMs by providing examples of a language and tasks to accomplish based on linguistic analysis (Bean et al., 2024; Tanzer et al., 2024). These problems require a mix of deductive, abductive, and analogical reasoning, allowing broad relevance beyond linguistics (Bean et al., 2024). Linguistics Olympiads are a common source of these problems, where puzzles have been curated for student competitions, ensuring the quality of the puzzles as well as their solvability by someone with no prior knowledge of the languages in question (Şahin et al., 2020; Chi et al., 2024). Morphological complexity has been shown to be particularly challenging for LLMs, as well as languages with lower similarity to English (Choudhary et al., 2025). Advanced problems from these competitions typically draw on low-resource languages where LLMs have minimal pre-training exposure, and remain difficult for top models (Sánchez et al., 2024; Lian et al., 2025). However, models show signs of partial contamination even in low resource languages (Bean et al., 2024), indicating a need for further methods to improve benchmark validity.

3 LINGOLY-TOO BENCHMARK

LINGOLY-TOO is a reasoning benchmark extending the LINGOLY benchmark (Bean et al., 2024). It consists of 1,203 questions and a total of 6,995 sub-question and answer pairs, generated through an obfuscation process designed by experts to preserve the intrinsic reasoning steps needed to solve each question, and to generate novel test data that measure the generalisable reasoning abilities of models.

3.1 LINGUISTICS TASKS

The benchmark is adapted from 82 problems from the UKLO. Linguistics Olympiad problems have several desirable features. All problems are grounded in natural languages, but require no preliminary knowledge in linguistics or any other domain beyond what is expected from middle and high school students. Each problem is self-contained and can be solved from the provided context using general reasoning and pattern matching skills. Answers vary in format, including free response and multiple choice, and can be evaluated with simple rule-based matching. Problems span five difficulty levels (Breakthrough, Foundation, Intermediate, Advanced, Round 2) sourced from UKLO.[2]

A typical problem sheet (referred to hereafter as a *problem*) consists of multiple questions, each question consists of multiple *(sub-question, answer)* pairs. Each problem presents words or phrases in a language of focus (termed *Problemese*) alongside their translations into the language in which the problem is written (termed *Solverese*) (Bozhanov & Derzhanski, 2013). The solver should deduce the underlying patterns and structures and use them to produce a best answer for each sub-question in a manner that resembles few-shot learning (Figure 2).

3.2 REASONING-EQUIVARIANT PERMUTATION

Although the original 82 problems include low-resource languages, these languages are increasingly represented in pre-training datasets, allowing models to rely on memorisation or knowledge rather than reasoning. We address this challenge by generating *reasoning-equivariant permutations* that obfuscate the original languages while preserving the underlying language mechanisms and solution steps. In Linguistics Olympiads, *obfuscation* describes the process that prevents cheating by removing metadata such as language names, language families, and geographic information that could identify the Problemese language (Asia Pacific Linguistics Olympiad, 2024; United Kingdom Linguistics Olympiad, 2021). We extend this approach by obfuscating the Problemese text itself, making it unlikely that models can solve problems using knowledge or memorisation.

Because the problems require symbolic reasoning over subwords (both morphemes such as suffixes, and phonemes), common permutation techniques that operate on complete words such as synonym replacement, paraphrasing, or word swapping are not suitable. These would corrupt the symbolic units and invalidate the problems. Instead, we manually created a ruleset for each problem to generate valid permutations of the targeted tokens (Problemese text). Valid permutations treat graphemes (both single letters and letter-combinations like English *th* and *sh*) as single units, and preserve any relationships between them which are relevant to solving the problem. They also preserve any loanwords or English cognates that might aid in solving the problem. Names of people, deities, and sacred places were also kept unchanged.

It is important to note that naive permutation of graphemes can easily render the problem unsolvable if it does not preserve the properties used in the underlying linguistic mechanism. We illustrate this with the example of minor vowel harmony highlighted in Problem 5 (Turkish, by Bozhidar Bozhanov). The suffix *-sVz* (meaning 'lacking', similar to the English '-less') can take four different forms depending on the previous vowel (Table 1). From a linguistic perspective, each of the four vowel pairs (*e*,*i*), (*o*,*u*), (*ö*,*ü*), (*a*,*ı*)

Last vowel	Suffix
e or *i*	*-siz*
o or *u*	*-suz*
ö or *ü*	*-süz*
a or *ı*	*-sız*

Table 1: **Problem 5 (Turkish) suffix form.** Suffix form of *-sVz* changes depending on the last vowel.

[2]For problems marked with multiple levels, we take the lower one. The distribution statistics of these difficulty levels can be found in Appendix D.

differ in the shape of the lips (*roundedness*) and position of the tongue (*backness*) when that sound is produced. For example, (*e*,*i*) are [−ROUND] and [−BACK], while (*o*,*u*) are [+ROUND] and [+BACK]. Any permutation which did not preserve these pairs would render the problem much harder and possibly unsolvable, since the vowel in the suffix would not correspond as naturally to the previous vowel. As such, all possible permutations must preserve the pairings, although the pairs may be permuted amongst themselves.

Our expert list of valid permutations takes into account the interactions between different segments to maintain two principles: (i) ensuring the permutations are reasoning-equivariant, and (ii) obtaining the maximum number of possible permutations. Further details can be found in Appendix B.

3.3 DATA GENERATION

We manually annotated 1,005 (sub-question, answer) pairs from 82 problems[3] with special tags for later processing. Metadata such as language names, language families, and geographic information, which had no impact on problem solvability but may provide clues for LLMs to use shortcuts were removed. A detailed description of the annotation process, including examples, is provided in Appendix C.

We randomly sampled up to 6 valid permutations per problem, and generated obfuscated versions of the problem by altering the Problemese text according to the selected permutations. Finally, we removed any special characters used for annotations. The original questions and the 6 permutations resulted in 6,995 (sub-question, answer) pairs across all problems.[4]

3.4 EVALUATION

Metrics We define a correct prediction as an exact match to the answer, with no credit given to partially correct answers. The correct answer often involves making small changes to words already present in the problem, so scores that offer partial credit may reward incorrect answers resulting from wrong reasoning.

Let $L(i, j, k, p)$ be the exact match score for problem $i \in \{1, ..., 82\}$, question $j \in \{1, \ldots, n_i\}$, sub-question $k \in \{1, \ldots, m_{i,j}\}$ after applying permutation $p \in \{0, \ldots, 6\}$, where n_i is the total number of questions in problem i and $m_{i,j}$ is the total number of sub-questions in problem i question j. To simplify notation, permutation $p = 0$ is the identity, i.e. the unpermuted case. We define $M_{obf}^{i,j}$ to be the average exact match score across all sub-questions in all (non-identity) permutations of problem i question j, that is, $M_{obf}^{i,j} = (1/(6m_{i,j})) \sum_{k=1}^{m_{i,j}} \sum_{p=1}^{6} L(i, j, k, p)$. In analyses where we compare results to those on the original (unpermuted) problems, we define $M_{og}^{i,j} = (1/m_{i,j}) \sum_{k=1}^{m_{i,j}} L(i, j, k, 0)$ to be the average exact match across all sub-questions in the unpermuted problem i question j. To evaluate a single model on the entire benchmark, we report the average across all problems and questions, that is

$$M_{obf} = \frac{1}{82} \sum_{i=1}^{82} \frac{1}{n_i} \sum_{j=1}^{n_i} M_{obf}^{i,j} \qquad \text{and} \qquad M_{og} = \frac{1}{82} \sum_{i=1}^{82} \frac{1}{n_i} \sum_{j=1}^{n_i} M_{og}^{i,j}. \tag{1}$$

Evaluation pipeline We broke down problems into individual questions and prompted models on each question with all of its (sub-question, answer) pairs. Each prompt consisted of a *preamble* (detailing background information about the language), *context* (key information the models need to reason over and infer the correct rules), all questions in the problem (some questions depend on previous ones) and the specific question to be answered. The prompt also specified the expected `json` structure response. The exact template used is provided in Appendix E. Since our aim is reasoning rather than instruction-following, we applied rule-based post-processing to recover improperly formatted responses. Remaining unparsed responses were counted as incorrect and detailed in Appendix G.

[3]We refer to problems by their numbering on the UKLO past problems database (United Kingdom Linguistics Olympiad, 2023).

[4]This is less than $7 \times 1,005 = 7,035$ as the total possible permutations for some problems are less than 6.

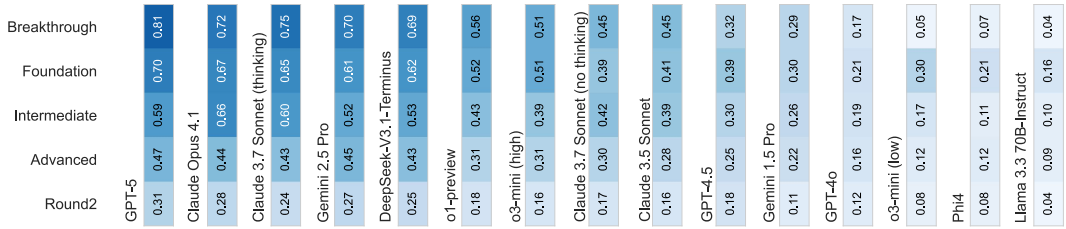

Figure 3: **Breakdown of M_{obf} scores by difficulty level.** For the stronger models, we see a clear decreasing trend in scores with increasing problem difficulty.

Models We evaluated fifteen reasoning and general-purpose model variations including open-source and closed-source models on the benchmark. See Appendix F for further details about the models.

3.5 Benchmark validation

Several measures were taken to ensure the quality and correctness of the benchmark. All problem annotations were performed manually and subsequently validated by two team members with expertise in Linguistics Olympiads. In addition, automated checks were incorporated to detect issues. Any identified errors triggered re-annotation and validation. A second round of both manual expert review and automated validation was conducted on the final data.

Obfuscation was designed to preserve reasoning steps and verified by team members with Linguistics Olympiads expertise. Additionally, two International Linguistics Olympiad (IOL) medallists, familiar with the original UKLO problems, audited a sample of obfuscations and independently confirmed that all problems remained solvable via the same reasoning steps.

Because obfuscated text no longer adhered to the original languages, it could impose a cognitive penalty in humans who may be more familiar with the original language orthographies. To estimate this effect, we ran a randomised controlled trial on a sample of six problems (chosen to be relatively easy to ensure novices had a realistic chance of solving them) with 172 human subjects. Participants were randomly assigned to either an original or obfuscated problem. Performance decreased 5.70% on obfuscated problems, likely due to their unfamiliarity, as noted by one auditor. Details of the study and analysis are provided in Appendix L.

4 Experiments and Analysis

4.1 Overall performance

Figure 1 shows all models' results on LingOly-TOO. The M_{og} metric (no obfuscation), is reported for comparison, which we posit is an over-estimation of reasoning abilities. We see that frontier models achieve ~ 0.59 on M_{og}. However, using M_{obf} metric (with obfuscation) , this drops to a maximum of 0.48. M_{obf} scores also reflect improved reasoning capabilities of reasoning models over general-purpose models. For example, GPT-5 scores higher than GPT-4.5, and Claude 3.7 (thinking) outperforms Claude 3.7 (no thinking) (0.44 vs. 0.30). In addition, o3-mini (high) outperforms o3-mini (low) by a larger margin (0.31 vs 0.13), indicating that increasing inference-time reasoning budget is useful for the benchmark tasks. Performance also seems related to the difficulty level in the best performing models but the pattern is not consistent in the weaker models (Figure 3). GPT-5 and Claude 3.7 Sonnet score highly (0.81 and 0.75, respectively) on *Breakthrough* problems but score below 0.31 at the *Round 2* level, demonstrating that the benchmark is far from saturated and offers a range of difficulties valuable for tracking progress. Overall, no model scores over 0.5, suggesting that despite recent breakthroughs, multi-hop inductive reasoning remain an open challenge.

	GPT-4o		Llama-3.3 70B	
Answer (% of total)	M_{og} (SE)	M_{obf} (SE)	M_{og} (SE)	M_{obf} (SE)
All	0.08 (0.00)	0.02 (0.00)	0.05 (0.00)	0.03 (0.00)
Digit (13.9%)	0.16 (0.01)	0.08 (0.00)	0.19 (0.01)	0.11 (0.01)
Single Char (14%)	0.03 (0.00)	0.03 (0.00)	0.05 (0.00)	0.04 (0.00)
Y/N (0.6%)	0.62 (0.04)	0.31 (0.04)	0.19 (0.04)	0.49 (0.04)
Other (71.6%)	0.07 (0.00)	0.01 (0.00)	0.02 (0.00)	0.01 (0.00)

Table 2: **Performance in the *no context* setting.** In the *no context* setting, prompts contain insufficient information to answer questions. Scores are categorised by answer type.

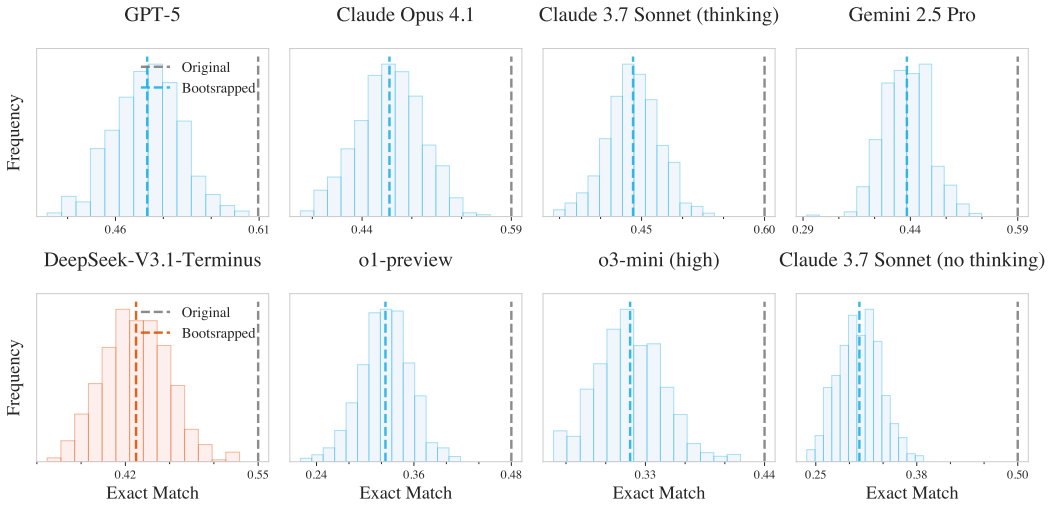

Figure 4: **Score distribution across bootstrapped samples.** Distribution of scores across 500 bootstrapped samples of our data by model. Each consists of 82 problems. Open source models are shown in orange while proprietary models are in blue (for full results, see Appendix I).

4.2 Knowledge vs. reasoning abilities

Ability to score without reasoning A key goal of the benchmark is to minimise the effects of models' knowledge on reasoning estimates. We evaluate LLMs' ability to score without reasoning on obfuscated data by adopting the *no context* setting from Bean et al. (2024), where critical information is removed from context, rendering the problems unsolvable by reasoning alone. We conduct a small experiment using 30 permutations per problem with two models (Llama 3.3 70B and GPT-4o).

Table 2 shows the results in the *no context* setting. M_{obf} for Llama 3.3 70B and GPT-4o is 0.02 and 0.03 respectively. When only considering questions with lower chance of random guessing (Other), M_{obf} further drops to ~ 0.01 in both models, demonstrating the effectiveness of obfuscation. This is not the case in the unobfuscated questions where GPT-4o is more successful in circumventing reasoning, scoring $M_{og} \sim 0.07$ through knowledge and memorisation. Inspecting a few examples where GPT-4o was successful reveals cases where GPT-4o correctly answers through direct translation, such as generating the correct Welsh translation: *Aeth meddyg i Gymru* to the English sentence *A doctor went to Wales*.

Reasoning gap To compare benchmark estimates with those from only the unobfuscated questions, we construct a bootstrap-style distribution over permutations. Specifically, we create 500 bootstrap sets, each containing all 82 problems. For each problem, we independently and uniformly select one of its seven versions (original plus six permutations) and compute the score. Figure 4 shows the distribution of scores on bootstrapped sets. When using unobfuscated problems, the score (M_{og}) consistently appears at the right tail of the distribution, reflecting a significant decrease in models'

Score	Standard	Dash	Character
Original	0.087	0.051	0.053
Permuted	0.050	0.045	0.035

Table 3: **Exact match score with varying tokenisation.** 'Dash' tokenisation inserts a dash between each character in the Problemese data. 'Character' tokenisation forces the model to tokenise each character individually.

performance with obfuscation. A similar gap is observed in the subset of problems used in the human study, where average model performance falls from 0.45 to 0.37 (8.6%) on obfuscated problems. However, reasoning models incur a smaller performance drop, on par with humans (5.8%), suggesting that reasoning models have reduced the gap in solving problems in their original orthography versus obfuscated versions compared with general-purpose models.

Tokenisation effect A possible explanation of the performance drop is that uncommon character sequences would affect LLMs through tokenisation by producing unuseful token representations. As tokenisers of LLMs are trained using frequency statistics on languages in their original orthography, sub-optimal tokenisation could explain the fall in model performance. We conduct a small experiment on a multilingual model (Aya-23-35B) to compare three variations of tokenisation: standard tokenisation, introducing a dash between each character in the Problemese and tokenising each character separately. Table 3 shows that enforcing separation of the tokens does not improve model performance. For both the original and obfuscated versions, exact match score does not improve if the tokenisation is altered, pointing towards explanations of failed reasoning other than the standard tokenisation. Additional tokenisation results are detailed in Appendix K.

Variance across permutations Models score higher on unobfuscated problems, possibly by utilising their knowledge from exposure to the original languages. Since obfuscation preserves solution logic, we quantify the performance difference due to obfuscation. For a fixed permutation p of problem i, we define $\Delta^{i,p}_{obf}$ as the difference between the average score across all questions of problem i under obfuscation p and the average score of all questions for the unobfuscated problem. That is

$$M^i_{og} = 1/n_i \sum_{j=1}^{n_i} M^{i,j}_{og}, \quad \text{and} \quad \Delta^{i,p}_{obf} = \frac{1}{n_i} \sum_{j=1}^{n_i} \left(\frac{1}{m_{i,j}} \sum_{k=1}^{m_{i,j}} L(i,j,k,p) \right) - M^i_{og}.$$

Then $\Delta^{i,p}_{obf}$ is negative when the model performs worse on a problem permutation, zero when there is no difference, and positive otherwise. Careful inspection reveals that although on average these differences are negative, model performance gaps vary across problems as well as across permutations within each individual problem. In a few cases such as Problem 24 (Tadaksahak) and Problem 67 (Navajo), some permutations even led to improved performance. We also see problems that had similar effects upon permutation across several models, indicating similarities in model failure modes. Appendix H includes full analysis of $\Delta^{i,p}_{obf}$ across all generated permutations of the 82 problems.

Language resourcedness effect Problems from higher resource languages generally show large negative $\Delta^i_{obf} = 1/6 \sum_p \Delta^{i,p}_{obf}$ values across all models such as Problem 74 (Japanese): -0.59, Problem 76 (Finnish): -0.59 and Problem 178 (Italian): -0.57. We approximate language resourcedness using the number of speakers of the language (in logarithmic scale). To quantify its effect on the gap in reasoning estimates, we apply linear regression to predict Δ^i_{obf} from resourcedness, separately for each problem difficulty and for models grouped into reasoning and general-purpose categories. We apply the Bonferroni correction to account for multiple-testing when reporting significance.

Results (Figure 5) show that language resourcedness is negatively correlated with reasoning performance gaps at the higher difficulty levels. In reasoning and general-purpose models, coefficients are negative ($\beta < 0$, $p < 0.01$ for Advanced and Round 2 problems). One possible explanation is that when models are unable to reason at higher difficulty levels, models are better at guessing the answer in languages with larger amount of data available for training.

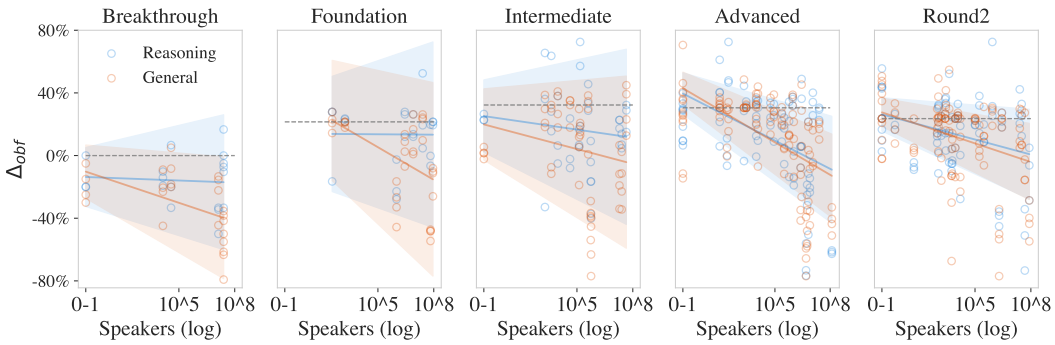

Figure 5: **The effect of permutation is larger for high-resource languages.** Each point represents all problems associated with a language of specific numbers of speaker. Solid lines are fitted regressions and shaded areas are the 95% confidence intervals.

4.3 Evidence of nascent reasoning

Let us define

$$M^i_{rob} = \frac{1}{n_i}\sum_{j=1}^{n_i}\frac{1}{m_{i,j}}\min_p\sum_{k=1}^{m_{i,j}} L(i,j,k,p) \qquad M_{rob} = \frac{1}{82}\sum_{i=1}^{82} M^i_{rob},$$

as a more robust metric of average model performance across all problems since it measures the minimum score achieved on any problem permutation. Under this metric, performance drops in all models (see Figure 6 for more details). The best model, GPT-5, only achieves 0.29 (a decrease of over 0.19). Reasoning models, which score the highest, are more consistent than general purpose models. An attempt for systematic analysis of reasoning traces using LLMs was unsuccessful as LLMs were unable to identify erroneous reasoning steps, making both type I and type II errors. We manually inspected the reasoning traces of Claude 3.7 (thinking) on two randomly selected problems where the model correctly answered questions in all permutations. We found examples where the model can identify subwords and suffixes across all permutations, infer their meaning, and deduce the correct answers (e.g. excerpt from one trace: *The verb prefixes seem to agree with the subject: "si-" for "they"*). However, these positive examples of a generalisable reasoning ability were among many more failed cases. Among them are traces with repeated analyses, sometimes drawing the same conclusions, others drawing different ones, even within the same permutation. More generally, M^i_{rob} shows that all models perform noticeably poorly, underscoring that reasoning is brittle and remains an open challenge to frontier models.

Reasoning with expert guidance By prompting with guidance steps written by team members based on the original problems, we evaluated whether model performance on the permuted problems improved. Using two problems chosen to have concise solutions (Problems 69 and 164) and four models from several providers (Claude 3.7 (thinking), Claude 3.5, Gemini 1.5 Pro, o3-mini high), we find that all models improve with the mean score on permuted questions (M_{obf}) increasing (0.66 to 0.76). This experiment suggests that, as in domains such as mathematics, multi-hop reasoning on unfamiliar language problems benefits from improved intermediate inductive reasoning steps.

4.4 Knowledge vs memorisation

While we assume a possibility that models are exposed to some of the unpermuted problems during training, we do not think that this accounts for total performance gaps. To illustrate, we were able to access and evaluate a subset of models on 5 problems from the UKLO 2025 that had not yet been published online at the time of the experiment and are of a higher difficulty level. Results in Table 9 show a comparable performance drop between the unpermuted and permuted cases to that observed in the benchmark. The difference in scores illustrates that the performance decrease is not simply due to memorising answers since these problems have not been included in models training. The gap also highlights the effectiveness of our permutation to control for knowledge even on unseen problems.

5 LIMITATIONS

Exact Match uniformly penalises all incorrect answers, including partially correct ones, and limits insights into failure modes. Yet, metrics such as BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004) are unsuitable for the very short answers common in our benchmark (note from Table 2 that 14% of our answers are single digits) and chrF (Popović, 2015) is sensitive to repeating related words from the context, which is a common behaviour in the models being tested. For example, Problem 16 4.1 (3) requires a translation of "your (plural) iguana" to Ulwa. The word for iguana in Ulwa is kahma, which is already given in the context. The correct answer is kah**mana**ma, as your (plural) is -mana-. Claude Opus 4.1 returned kahma**na**, which is the wrong form and placement of the possessive. If we assigned partial credit through e.g. edit distance, the answer from Claude will score highly, even though this is an incorrect answer based on incorrect reasoning. Assigning partial credit would unnecessarily inflate the baseline through repeating parts of words in the context. Future work could explore alternative evaluation metrics that capture partial solutions or help discern partially correct answers to ensure construct validity (Bean et al., 2025). Novel evaluation scores could credit correct intermediate reasoning steps alongside the final answers.

While problems in LINGOLY-TOO do not depend on any specific domain knowledge, we limit this work to inductive and deductive reasoning in the natural language modality. The benchmark is not a comprehensive evaluation of all reasoning abilities across all modalities but results suggest that improvements in certain reasoning abilities would translate to improvements in others.

6 CONCLUSION

We introduced LINGOLY-TOO, a benchmark for evaluating reasoning in language tasks. By applying expert-designed obfuscations to Linguistics Olympiad problems, we generated permutations that preserved solution logic while remaining novel to models. Our results showed that the benchmark effectively controls for model knowledge and memorisation, revealing that LLMs (including reasoning models) often rely on shortcuts rather than genuine reasoning. We further quantified how language resourcedness amplify these shortcuts. Finally, we observed that reasoning models were better able to apply symbolic and inductive reasoning and that progress reported in domains such as mathematics and coding appears to partially translate to language problems.

Our benchmark remains unsaturated, especially at the highest problem difficulty, leaving ample headroom. Our results support existing claims of model reliance on shortcuts, which biases reasoning ability estimates. Compared to the unobfuscated problems, deriving estimates from obfuscated problems paints a more conservative picture of the reasoning ability of frontier models. We find that reasoning consistency and robustness remain limited in frontier models and warrant further attention.

ETHICS STATEMENT

This paper aims to advance the field of machine learning. Constructing and publishing problems using low-resource languages to create a benchmark may lead to concerns about the exploitation of or other harm caused to the communities who use and own these languages (Tanzer et al., 2024). The problems in this benchmark are created either by native speakers or using published resources such as sketch grammars. In these cases, the relevant language communities have already given consent for a linguist to publish information on the language. In this paper, we transform existing puzzles, and do not create new content in the languages.

The obfuscations of the problem amount to a new orthography for the language, and do not change any other aspect of the language such as grammar or the meanings of words; do not alter the names of people, deities, or sacred places; and are not publicly available. All participants in the human study were provided with information about the language they had studied and its speakers after completing the study.

REPRODUCIBILITY STATEMENT

We release complete anonymised code and evaluation scripts for our benchmark for reproducibility at https://github.com/jkhouja/LingOly-TOO.